

YILAN ZHANG

King Abdullah University of Science and Technology, Saudi Arabia

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RESEARCH INTERESTS

Medical Image Analysis, Multimodal Learning, Computational Pathology

EDUCATION

PhD Student in KAUST

Mar 2024 – now

Computer Science

Master in Beihang University

Sep 2021 – Dec 2023

Control Science and Engineering, Ranking: 3/140, GPA:3.84(4)

Bachelor in Beihang University

Sep 2017 – Jun 2021

Control Science and Engineering, Economics (second bachelor's degree) Ranking: 7/83, GPA:3.76(4)

SELECTED PUBLICATIONS (ONLY FIRST AUTHOR)

Prototypical Information Bottlenecking and Disentangling for Multimodal Cancer Survival Prediction

The Twelfth International Conference on Learning Representations (ICLR) Spotlight, 2024

Yilan Zhang, Yingxue Xu, Jianqi Chen, Fengying Xie, and Hao Chen.

ECL: Class-Enhancement Contrastive Learning for Long-tailed Skin Lesion Classification

International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023, STAR Award

Yilan Zhang, Jianqi Chen, Ke Wang and Fengying Xie

TFormer: A throughout fusion transformer for multi-modal skin lesion diagnosis

Computers in Biology and Medicine (CBM), 2023

Yilan Zhang, Fengying Xie and Jianqi Chen

Dermoscopic image retrieval based on rotation-invariance deep hashing

Medical Image Analysis (MedIA), 2022

Yilan Zhang, Fengying Xie, Xuedong Song, Yushan Zheng, Jie Liu, and Juncheng Wang

A rotation meanout network with invariance for dermoscopy image classification and retrieval

Computers in Biology and Medicine (CBM), 2022

Yilan Zhang, Fengying Xie, Xuedong Song, Hangning Zhou, Yiguang Yang, Haopeng Zhang, and Jie Liu

PROJECTS & RESEARCH

Research Intern in HKUST SMART Lab

May 2023 – Oct 2023

Advisor: Prof. Hao Chen

Research Intern in Biomap

Nov 2022 – May 2023

Biological Image Group

Research on Dermoscopy Image Classification and Retrieval System

Nov 2020 – Mar 2023

Collaborating with Peking Union Medical College

HONORS

Outstanding Graduates of Beijing Municipality	Dec, 2023
National Scholarship, Ministry of Education of China	Sep 2022
Finalist Winners of the International Mathematical Contest in Modeling (MCM), COMAP	May 2020

FULL PUBLICATIONS

Structural Prognostic Event Modeling for Multimodal Cancer Survival Analysis

arXiv preprint arXiv:2512.01116, 2025

Yilan Zhang, Nanbo Li, Changchun Yang, Jürgen Schmidhuber and Xin Gao.

Prototypical Information Bottlenecking and Disentangling for Multimodal Cancer Survival Prediction

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A rotation meanout network with invariance for dermoscopy image classification and retrieval

Computers in Biology and Medicine (CBM), 2022

Yilan Zhang, Fengying Xie, Xuedong Song, Hangning Zhou, Yiguang Yang, Haopeng Zhang, and Jie Liu

Early Diagnosis Model of Mycosis Fungoides Based on Intelligent Analysis of Dermoscopic Images

Medical Journal Of Peking Union Medical College Hospital (Rhzh Test), 2021

Zhaorui Liu, Yilan Zhang (Co-first author), Fengying Xie, and Jie Liu

Early diagnosis model of mycosis fungoides and five inflammatory skin diseases based on multi-modal data-based convolutional neural network

British Journal of Dermatology, 2025

Zhaorui Liu, Yilan Zhang (Co-first author), Ke Wang, Fengying Xie, Jie Liu

Improving Representation of High-frequency Components for Medical Visual Foundation Models

IEEE Transactions on Medical Imaging, 2025

Yuetan Chu, Yilan Zhang (Co-first author), Zhongyi Han, Changchun Yang, Longxi Zhou, Gongning Luo, Chao Huang, Xin Gao

Beyond benchmarks: the promise of multimodal artificial intelligence in real-world dermatology

British Journal of Dermatology, 2025

Zhaorui Liu, Yilan Zhang (Co-first author), Ke Wang, Fengying Xie, Jie Liu

Enhancing Diagnosis of Psoriasis and Inflammatory Skin Diseases: A Spatially Aligned Multimodal Model Integrating Clinical and Dermoscopic Images

Journal of Investigative Dermatology, 2025

Juncheng Wang, Yilan Zhang (Co-first author), Fengying Xie, Jie Liu

An Inclusive Foundation Model for Generalizable Cytogenetics in Precision Oncology

arXiv preprint arXiv:2505.15868, 2025

Changchun Yang, Weiqian Dai, Yilan Zhang (Co-first author), Siyuan Chen, Jingdong Hu, Junkai Su, Yuxuan Chen, Ao Xu, Na Li, Xin Gao, Yongguo Yu

Skin lesion classification based on two-modal images using a multi-scale fully-shared fusion network

Computer Methods and Programs in Biomedicine, 2023

Yiguang Yang, Fengying Xie, Haopeng Zhang, Juncheng Wang, Jie Liu, Yilan Zhang, and Haidong Ding

NoTAC: A Noise-Tolerance Automatic Cleaning Framework for Bone Marrow Karyotyping Data

IEEE journal of biomedical and health informatics, 2025

Rihan Huang, Siyuan Chen, Yafei Li, Chunling Zhang, Yilan Zhang, Changchun Yang, Na Li, Jingdong Hu, Ao Xu, Junkai Su, Xin Gao, Huidan Li, Jiatao Lou

Past: A multimodal single-cell foundation model for histopathology and spatial transcriptomics in cancer

arXiv preprint arXiv:2507.06418, 2025

Changchun Yang, Haoyang Li, Yushuai Wu, Yilan Zhang, Yifeng Jiao, Yu Zhang, Rihan Huang, Yuan Cheng, Yuan Qi, Xin Guo

I2M2Net: Inter/Intra-modal Feature Masking Self-distillation for Incomplete Multimodal Skin Lesion Diagnosis

International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), Workshop, 2024

Ke Wang, Linwei Qiu, Yilan Zhang, Fengying Xie

Early mycosis fungoides identification based on multimodal image fusion.

Journal of Data Acquisition Processing, 2023

Fengying Xie, Danpei Zhao, Ke Wang, Zhaorui Liu, Yukun Wang, Yilan Zhang, Jie Liu

Digital-to-Physical visual consistency optimization for adversarial patch generation in remote sensing scenes

IEEE Transactions on Geoscience and Remote Sensing, 2024

Jianqi Chen, Yilan Zhang, Chenyang Liu, Keyan Chen, Zhengxia Zou, Zhenwei Shi

Dense pixel-to-pixel harmonization via continuous image representation

IEEE Transactions on Circuits and Systems for Video Technology, 2023

Jianqi Chen, Yilan Zhang, Zhengxia Zou, Keyan Chen, Zhenwei Shi

Diffusion models for imperceptible and transferable adversarial attack

IEEE Transactions on Pattern Analysis and Machine Intelligence, 2024

Jianqi Chen, Hao Chen, Keyan Chen, Yilan Zhang, Zhengxia Zou, Zhenwei Shi

Zero-shot image harmonization with generative model prior

IEEE Transactions on Multimedia, 2025

Jianqi Chen, Yilan Zhang, Zhengxia Zou, Keyan Chen, Zhenwei Shi